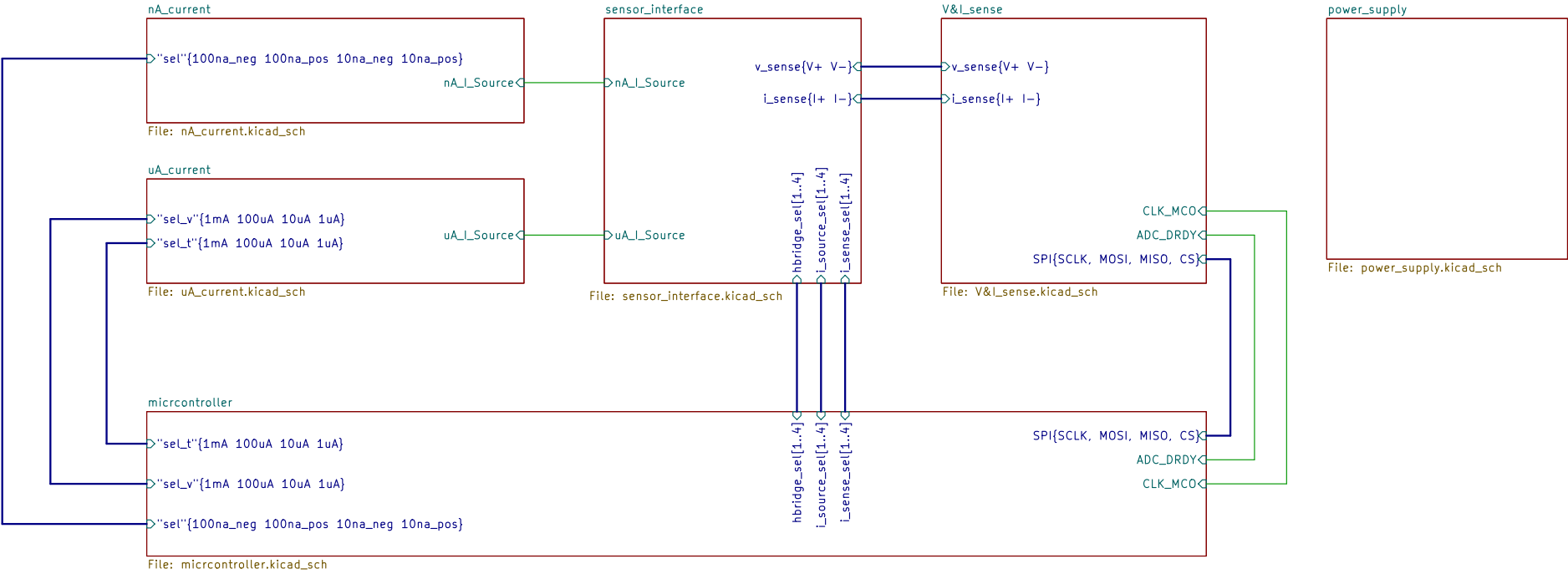


OVERVIEW (PAGE 1)



Sheet: /
File: Black_Body_Temperature_Sensor_PCB.kicad_sch

Title:

Size: A4

Date:

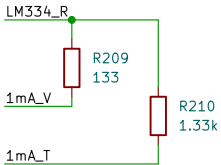
KiCad E.D.A. 10.0.2

Rev:

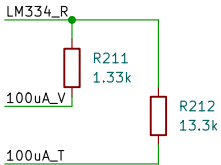
Id: 1/7

uA Current (PAGE 2)

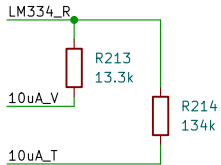
1mA current source selection



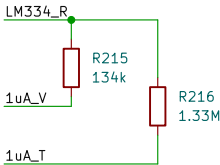
100uA current source selection



10uA current source selection

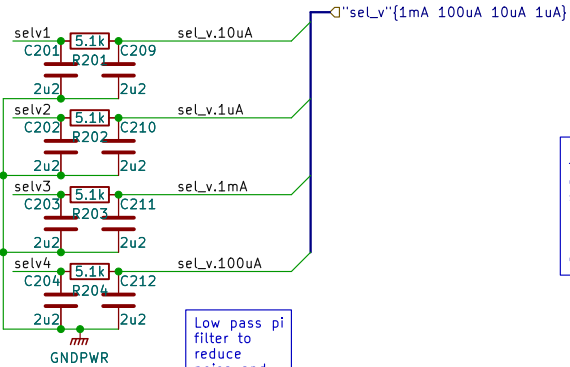


1uA current source selection



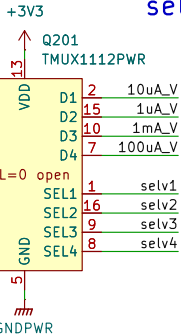
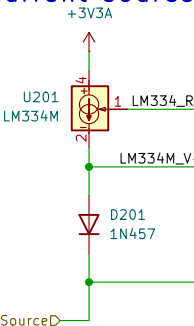
$I_{set} = 0.134/R1$
(R1 is between LM334_R and 1uA_V)
 $R2 = 10R1$

Current output selection

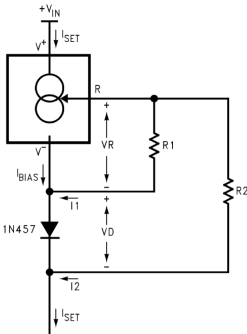
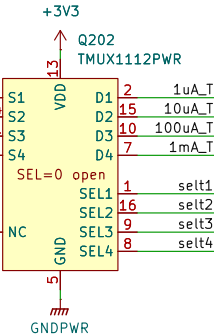


current source

LM334M is a temperature dependant current source which can be stabilized using external resistors and a diode



TMUX is a very low noise switch to select different resistances to change current output



Sheet: /uA_current/
File: uA_current.kicad_sch

Title:

Size: A4

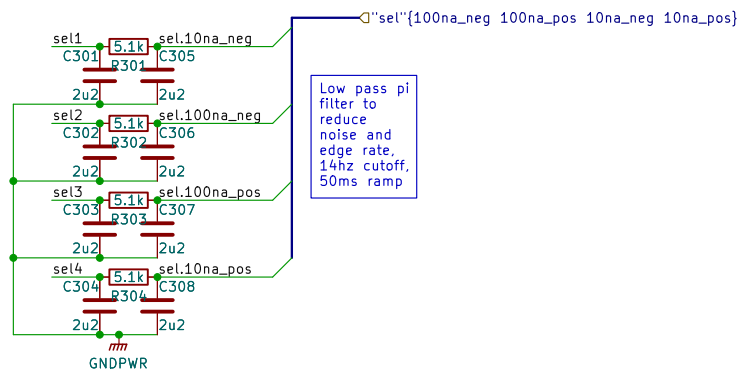
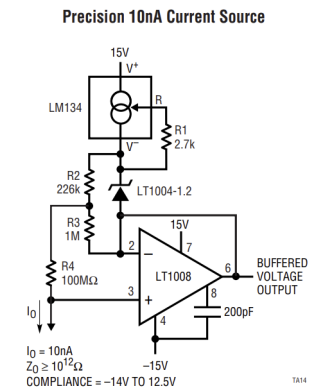
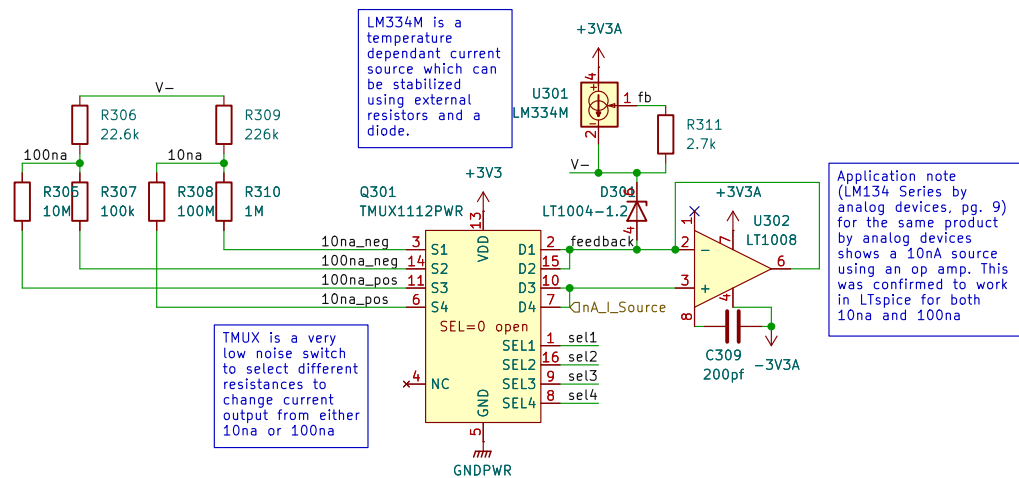
Date:

KiCad E.D.A. 10.0.2

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Id: 2/7

nA Current (PAGE 3)



Sheet: /nA_current/
File: nA_current.kicad_sch

Title:

Size: A4

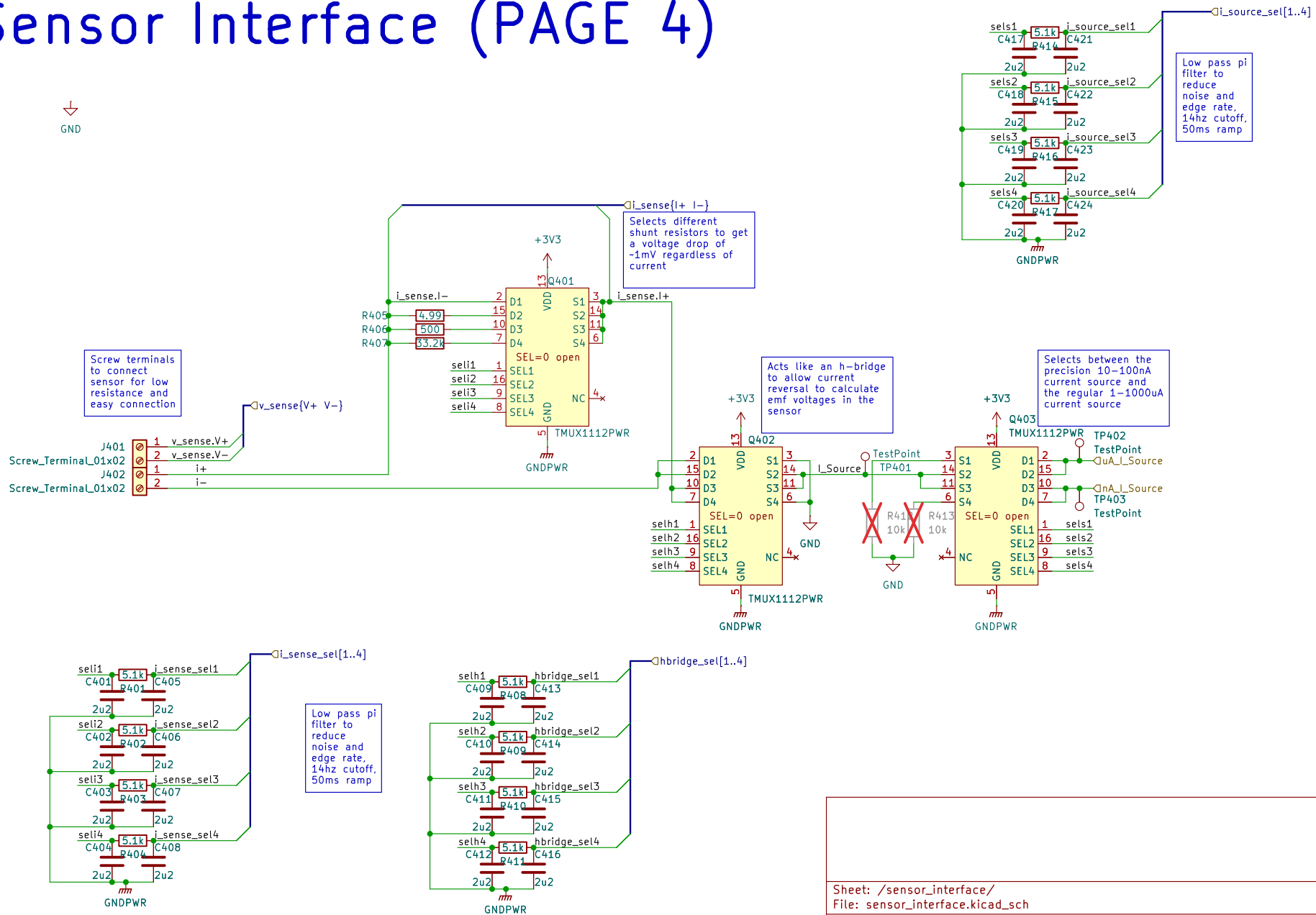
Date:

KiCad E.D.A. 10.0.2

Rev:

Id: 3/7

Sensor Interface (PAGE 4)



Sheet: /sensor_interface/
File: sensor_interface.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.2

Rev:

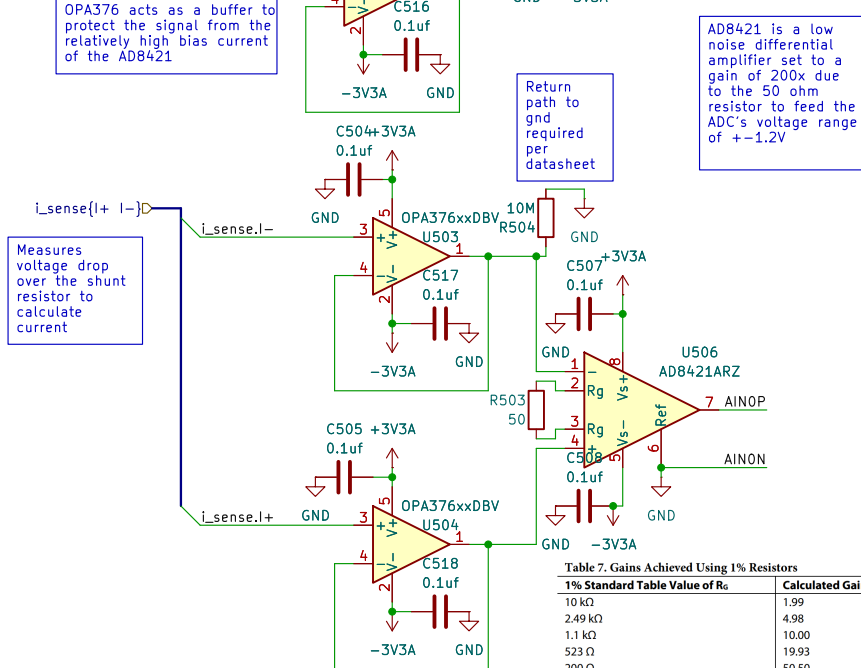
Id: 4/7

Measures voltage drop over the temperature probe

Return path to gnd required per datasheet

sense_iso_V+

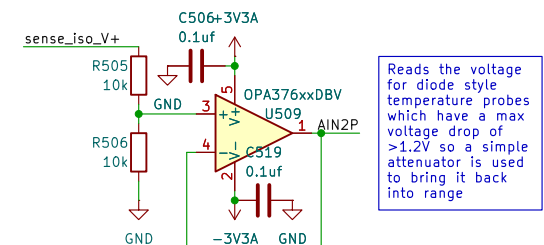
sense_iso_V-



1% Standard Table Value of R_0	Calculated Gain
10 k Ω	1.99
2.49 k Ω	4.98
1.1 k Ω	10.00
523 Ω	19.93
200 Ω	50.50
100 Ω	100.0
49.9 Ω	199.4
20 Ω	496.0
10 Ω	991.0
4.99 Ω	1985



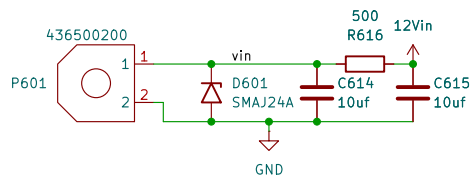
	GR-300-AA		
	R(Ω)	dR/dT (Ω /K)	(T/R)· (dR/dT)
0.05 K	—	—	—
0.1 K	—	—	—
0.2 K	—	—	—
0.3 K	35180	-512200	-4.367
0.5 K	5443	-34800	-3.197
1.0 K	875.7	-1901	-2.170
1.4 K	448.6	-581.3	-1.814
2.0 K	248.8	-187.4	-1.507
4.2 K	94.46	-26.56	-1.181
10 K	33.20	-3.965	-1.194
40 K	7.789	-0.2351	-1.207
77.4 K	3.504	-0.0496	-1.096
100 K	2.716	-0.0238	-0.8754



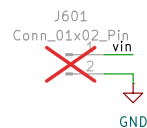
Rev:
Id: 5/7

Power Supply (PAGE 6)

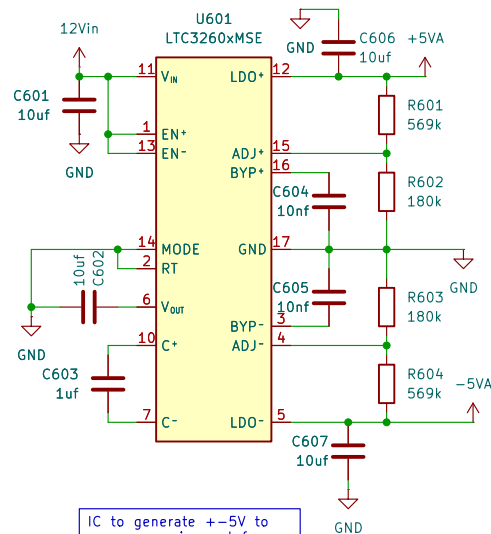
Voltage In



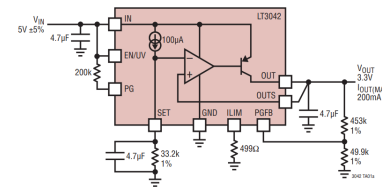
Solderable power connector



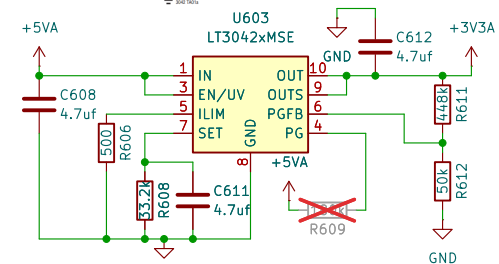
+12V in -> +5V & -5V
100uV noise



IC to generate +/-5V to remove requirement for expensive power supply

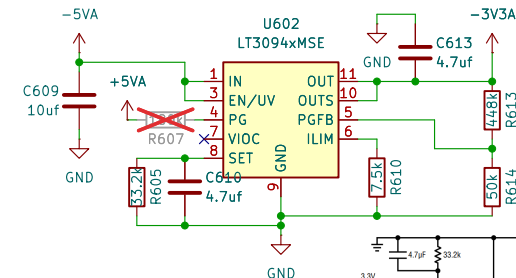


+5V in -> +3V3
0.8uV noise

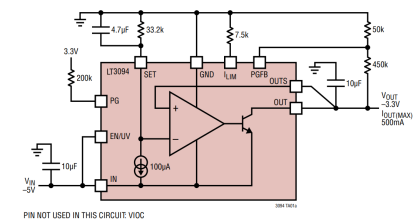
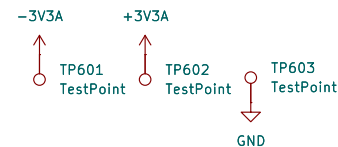


Voltage cutoff
 $0.3 * (1+450k/50k) = 3V$

-5V in -> -3V3
0.8uV noise



Voltage cutoff
 $0.3 * (1+450k/50k) = 3V$



PIN NOT USED IN THIS CIRCUIT VIOC

Sheet: /power_supply/
File: power_supply.kicad_sch

Title:

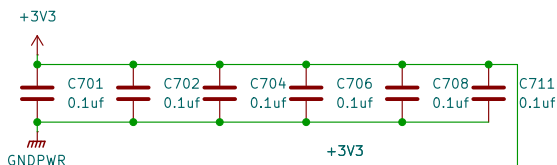
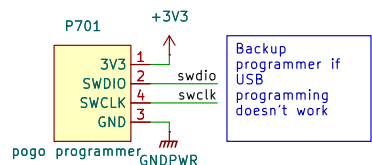
Size: A4
KiCad E.D.A. 10.0.2

Date:

Rev:
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Microcontroller (PAGE 7)

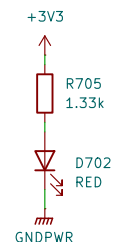
st-link programmer



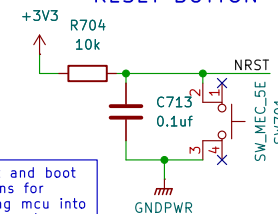
28pf calculated using ecsxtal.com with 18pf load capacitance (from datasheet) and 4pf stray capacitance



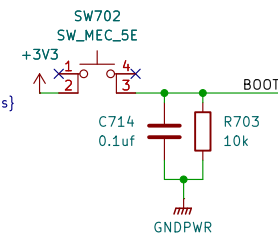
POWER INDICATOR



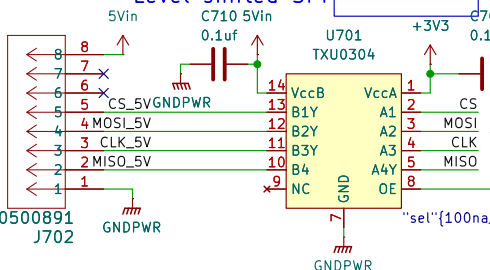
RESET BUTTON



USB PROGRAM BUTTON

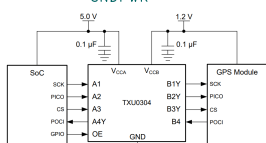
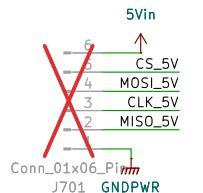


Level shifted SPI



5040500891 J702

Solderable connector

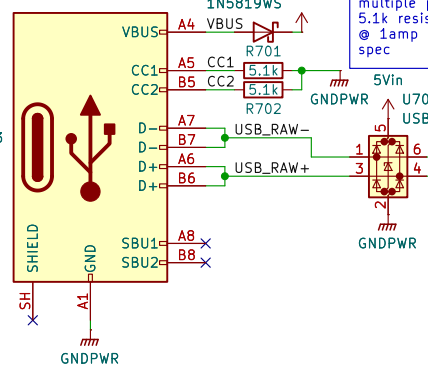


SPI{SCLK, MOSI, MISO, CS}

"sel_v"{1mA 100uA 10uA 1uA}

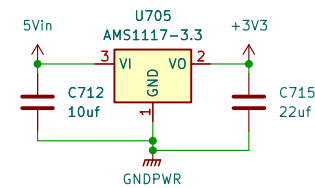
i_source_sel{1..4}

USB-C



USB_C_Receptacle_USB2.0_16P

Microcontroller 3.3V



Sheet: /micrcontroller/
File: micrcontroller.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.2

Date:

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